Figure 2.14e are the same, although the position of the outflow hydrograph in time is modified due to the smoothing of the surface runoff and the channel processes.

The processes that have been discussed in the previous sections all act simultaneously to transform the incoming rainfall from that shown in Figure 2.14a to the corresponding outflow hydrograph of Figure 2.14f. This example serves to illustrate the runoff process for a small local area. If the watershed is of appreciable size or if the storm is large, areal and time variations and other factors add a new level of complexity to the problem.

2.6 TRAVEL TIME

The travel time of runoff is very important in hydrologic design. In the design of inlets and pipe drainage systems, travel times of surface runoff must be estimated. Some peak discharge methods (Chapter 5) use the time of concentration as input to obtain rainfall intensities from the intensity-duration-frequency curves. Hydrograph times-to-peak, which are in some cases computed from times of concentration, are used with hydrograph methods (Chapter 6). Channel routing methods (Chapter 7) use computed travel times in routing hydrographs through channel reaches. Thus, estimating travel times are central to a variety of hydrologic design problems.

2.6.1 Time of Concentration

The time of concentration, which is denoted as t_c, is defined as the time required for a particle of water to flow from the hydraulically most distant point in the watershed to the outlet or design point. Factors that affect the time of concentration are the length of flow, the slope of the flow path, and the roughness of the flow path. For flow at the upper reaches of a watershed, rainfall characteristics, most notably the intensity, may also influence the velocity of the runoff.

Various methods can be used to estimate the time of concentration of a watershed. When selecting a method to use in design, it is important to select a method that is appropriate for the flow path. Some estimation methods were designed and can be classified as “lumped” in that they were designed and calibrated to be used for an entire watershed; the SCS lag formula is an example of this method. These methods have t_c as the dependent variable. Other methods are intended for one segment of the principal flow path and produce a flow velocity that can be used with the length of that segment of the flow path to compute the travel time on that segment. With this method, the time of concentration equals the sum of the travel times on each segment of the principal flow path.

In classifying these methods so that the proper method can be selected, it is useful to describe the segments of flow paths. Sheet flow occurs in the upper reaches of a watershed. Such flow occurs over short distances and at shallow depths prior to the point where topography and surface characteristics cause the flow to concentrate in rills and swales. The depth of such flow is usually 20 to 30 mm (0.8 to 1.2 in) or less. Concentrated flow is runoff that occurs in rills and swales and has depths on the order of 40 to 100 mm (1.6 to 3.9 in). Part of the principal flow path may include pipes or small streams. The travel time through these segments would be computed separately. Velocities in open channels are usually determined assuming bank-full depths.

2.6.2 Velocity Method

The velocity method (sometimes referred to as the segment method) can be used to estimate travel times for sheet flow, shallow concentrated flow, pipe flow, or channel flow. It is based on estimating the travel time from the length and velocity:

$$T_t = \frac{L}{60V} \quad (2.3)$$

where,

T_t = travel time, min
L = flow length, m (ft)
V = flow velocity, m/s (ft/s).

The travel time is computed for the principal flow path. When the principal flow path consists of segments that have different slopes or land covers, the principal flow path should be divided into segments and Equation 2.3 used for each flow segment. The time of concentration is then the sum of travel times:

$$t_c = \sum_{i=1}^{k} T_{t_i} = \sum_{i=1}^{k} \left(\frac{L_i}{60 V_i} \right) \quad (2.4)$$

where,

k = number of segments
i = subscript referring to each flow segment.

Velocity is a function of the type of flow (overland, sheet, rill and gully flow, channel flow, pipe flow), the roughness of the flow path, and the slope of the flow path. Some methods also include a rainfall index such as the 2-year, 24-hour rainfall depth. A number of methods have been developed for estimating the velocity.

2.6.2.1 Sheet-Flow Travel Time

Sheet flow is a shallow mass of runoff on a plane surface with the depth uniform across the sloping surface. Typically flow depths will not exceed 50 mm (2 in). Such flow occurs over relatively short distances, rarely more than about 90 m (300 ft), but most likely less than 25 m (80 ft). Sheet flow rates are commonly estimated using a version of the kinematic wave equation. The original form of the kinematic wave time of concentration is:

$$t_c = \frac{\alpha}{i^{0.4}} \left(\frac{nL}{\sqrt{S}} \right)^{0.6} \quad (2.5)$$

where,

t_c = time of concentration, min
n = roughness coefficient (see Table 2.1)
L = flow length, m (ft)
i = rainfall intensity, mm/h (in/h), for a storm that has a return period T and duration of t_c minutes
S = slope of the surface, m/m (ft/ft)
α = unit conversion constant equal to 6.9 in SI units and 0.93 in CU units.

Some hydrologic design methods, such as the rational equation, assume that the storm duration equals the time of concentration. Thus, the time of concentration is entered into the IDF curve to find the design intensity. However, for Equation 2.5, i depends on t_c and t_c is not initially known. Therefore, the computation of t_c is an iterative process. An initial estimate of t_c is assumed and used to obtain i from the intensity-duration-frequency curve for the locality. The t_c is computed

2.6.2.2 Shallow Concentrated Flow

After short distances, sheet flow tends to concentrate in rills and then gullies of increasing proportions. Such flow is usually referred to as shallow concentrated flow. The velocity of such flow can be estimated using an empirical relationship between the velocity and the slope:

$$V = \alpha k S^{0.5} \qquad (2.7)$$

where,

V = velocity, m/s (ft/s)
S = slope, m/m (ft/ft)
k = dimensionless function of land cover (see Table 2.2)
α = unit conversion constant equal to 10 in SI and 33 in CU units.

Table 2.2. Intercept Coefficients for Velocity vs. Slope Relationship (McCuen, 1989)

k	Land Cover/Flow Regime
0.076	Forest with heavy ground litter; hay meadow (overland flow)
0.152	Trash fallow or minimum tillage cultivation; contour or strip cropped; woodland (overland flow)
0.213	Short grass pasture (overland flow)
0.274	Cultivated straight row (overland flow)
0.305	Nearly bare and untilled (overland flow); alluvial fans in western mountain regions
0.457	Grassed waterway (shallow concentrated flow)
0.491	Unpaved (shallow concentrated flow)
0.619	Paved area (shallow concentrated flow); small upland gullies

2.6.2.3 Pipe and Channel Flow

Flow in gullies empties into channels or pipes. In many cases, the transition between shallow concentrated flow and open channels may be assumed to occur where either the blue-line stream is depicted on USGS quadrangle sheets (scale equals 1:24000) or when the channel is visible on aerial photographs. Channel lengths may be measured directly from the map or scale photograph. However, depending on the scale of the map and the sinuosity of the channel, a map-derived channel length may be an underestimate. Pipe lengths should be taken from as-built drawings for existing systems and design plans for future systems.

Cross-section information (i.e., depth-area and roughness) can be obtained for any channel reach in the watershed. Manning's equation can be used to estimate average flow velocities in pipes and open channels:

$$V = \frac{\alpha}{n} R^{2/3} S^{1/2} \qquad (2.8)$$

where,

V = velocity, m/s (ft/s)
n = Manning's roughness coefficient
R = hydraulic radius, m (ft)

The unit peak discharge is computed with Equation 5.26 by interpolating c_0 , c_1, and c_2 from Table 5.5 using a type II distribution. The peak discharge is also calculated as follows.

Variable	SI Unit	CU Unit
$q_u = 10^{2.5444 - 0.61587\ \log(0.26) - 0.15928[\log(0.26)]}$	$= (0.000431)\ 10^{2.85}$ $= 0.305\ m^3/s/km^2/mm$	$= (1)\ 10^{2.85}$ $= 708\ ft^3/s/mi^2/in$
$q_p = q_u AQ$	$= 0.305\ (0.176\ km^2)(62\ mm)$ $= 3.3\ m^3/s$	$= 708\ (0.068\ mi^2)\ (2.46\ in)$ $= 120\ ft^3/s$

5.3 RATIONAL METHOD

One of the most commonly used equations for the calculation of peak discharges from small areas is the rational formula. The rational formula is given as:

$$Q = \frac{1}{\alpha}\ C\ i\ A \qquad (5.28)$$

where,

Q = the peak flow, m³/s (ft^3/s)
i = the rainfall intensity for the design storm, mm/h (in/h)
A = the drainage area, ha (acres)
C = dimensionless runoff coefficient assumed to be a function of the cover of the watershed and often the frequency of the flood being estimated
α = unit conversion constant equal to 360 in SI units and 1 in CU units.

5.3.1 Assumptions

The assumptions in the rational formula are as follows:

1. The drainage area should be smaller than 80 hectares (200 acres).

2. The peak discharge occurs when the entire watershed is contributing.

3. A storm that has a duration equal to t_c produces the highest peak discharge for this frequency.

4. The rainfall intensity is uniform over a storm time duration equal to the time of concentration, t_c. The time of concentration is the time required for water to travel from the hydrologically most remote point of the basin to the outlet or point of interest.

5. The frequency of the computed peak flow is equal to the frequency of the rainfall intensity. In other words, the 10-year rainfall intensity, i, is assumed to produce the 10-year peak discharge.

5.3.2 Estimating Input Requirements

The runoff coefficient, C, is a function of ground cover. Some tables of C provide for variation due to slope, soil, and the return period of the design discharge. Actually, C is a volumetric coefficient that relates the peak discharge to the "theoretical peak" or 100 percent runoff, occurring when runoff matches the net rain rate. Hence C is also a function of infiltration and other hydrologic abstractions. Some typical values of C for the rational formula are given in

Table 5.7. Should the basin contain varying amounts of different covers, a weighted runoff coefficient for the entire basin can be determined as:

$$Weighted\ C = \frac{\sum C_i A_i}{A} \qquad (5.29)$$

where,

C_i = runoff coefficient for cover type i that covers area A_i

A = total area.

5.3.3 Check for Critical Design Condition

When the rational method is used to design multiple drainage elements (i.e. inlets and pipes), the design process proceeds from upstream to downstream. For each design element, a time of concentration is computed, the corresponding intensity determined, and the peak flow computed. For pipes that drain multiple flow paths, the longest time of concentration from all of the contributing areas must be determined. If upstream pipes exist, the travel times in these pipes must also be included in the calculation of time of concentration.

In most cases, especially as computations proceed downstream, the contributing area with the longer time of concentration also contributes the greatest flow. Taking the case of two contributing areas, as shown in Figure 5.3a, the longest time of concentration of the two areas is used to determine the time of concentration for the combined area. When the rainfall intensity corresponding to this time of concentration is applied to the rational equation, as shown below, for the combined area and runoff coefficient, the appropriate design discharge, Q, results.

$$Q = \frac{1}{\alpha}(C_1 A_1 + C_2 A_2)\, i_1 \qquad (5.30)$$

However, it may be possible for the larger contributing flows to be generated from the contributing area with a shorter time of concentration. If this occurs, it is also possible that, if the longer time of concentration is applied to the combined drainage area, the resulting design flow would be an underestimate. Therefore, a check for a critical design condition must be made.

$$Q' = \frac{1}{\alpha}(C_1 A_1 \frac{t_2}{t_1} + C_2 A_2)\, i_2 \qquad (5.31)$$

where,

Q' = design check discharge

t_1 = time of concentration for area 1

t_2 = time of concentration for area 2.

Table 5.7. Runoff Coefficients for Rational Formula (ASCE, 1960)

Type of Drainage Area	Runoff Coefficient
Business:	
Downtown area	0.70-0.95
Neighborhood areas	0.50-0.70
Residential:	
Single-family areas	0.30-0.50
Multi-units, detached	0.40-0.60
Multi-units, attached	0.60-0.75
Suburban	0.25-0.40
Apartment dwelling areas	0.50-0.70
Industrial:	
Light areas	0.50-0.80
Heavy areas	0.60-0.90
Parks, cemeteries	0.10-0.25
Playgrounds	0.20-0.40
Railroad yard areas	0.20-0.40
Unimproved areas	0.10-0.30
Lawns:	
Sandy soil, flat, < 2%	0.05-0.10
Sandy soil, average, 2 to 7%	0.10-0.15
Sandy soil, steep, > 7%	0.15-0.20
Heavy soil, flat, < 2%	0.13-0.17
Heavy soil, average 2 to 7%	0.18-0.22
Heavy soil, steep, > 7%	0.25-0.35
Streets:	
Asphalt	0.70-0.95
Concrete	0.80-0.95
Brick	0.70-0.85
Drives and walks	0.75-0.85
Roofs	0.75-0.95

If Q' > Q, Q' should be used for design; otherwise Q should be used. Equation 5.31 uses the rainfall intensity for the contributing area with the shorter time of concentration (area 2) and reduces the contribution of area 1 by the ratio of the times of concentration. This ratio approximates the fraction of the area that would contribute within the shorter duration. This is equivalent to reducing the contributing area as shown by the dashed line in Figure 5.4.

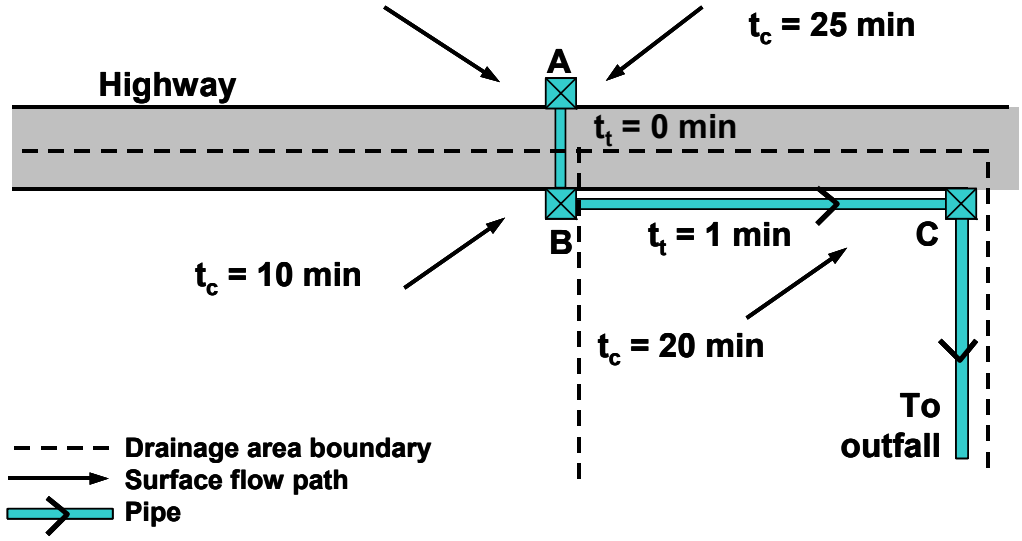


Figure 5.4. Storm drain system schematic

Example 5.5. A flooding problem exists along a farm road near Memphis, Tennessee. A low-water crossing is to be replaced by a culvert installation to improve road safety during rainstorms. The drainage area above the crossing is 43.7 ha (108 acres). The return period of the design storm is to be 25 years as determined by local authorities. The engineer must determine the maximum discharge that the culvert must pass for the indicated design storm.

The current land use consists of 21.8 ha (53.9 acres) of parkland, 1.5 ha (3.7 acres) of commercial property that is 100 percent impervious, and 20.4 ha (50.4 acres) of single-family residential housing. The principal flow path includes 90 m (295 ft) of short grass at 2 percent slope, 300 m (985 ft) of grassed waterway at 2 percent slope, and 650 m (2,130 ft) of grassed waterway at 1 percent slope. The following steps are used to compute the peak discharge with the rational method:

1. **Compute a Weighted Runoff Coefficient**: The tabular summary below uses runoff coefficients from Table 5.7. The average value is used for the parkland and the residential areas, but the highest value is used for the commercial property because it is completely impervious.

Description	C Value	SI Unit		CU Unit	
		Area (ha)	C_iA_i	Area (acres)	C_iA_i
Park	0.20	21.8	4.36	53.9	10.8
Commercial (100% impervious)	0.95	1.5	1.43	3.7	3.5
Single-family	0.40	20.4	8.16	50.4	20.2
Total		43.7	13.95	108.0	34.5

Equation 5.29 is used to compute the weighted C:

Variable	Value in SI	Value in CU
$Weighted\ C = \frac{\sum C_i A_i}{A}$	$= \frac{13.95}{43.7} = 0.32$	$= \frac{34.5}{108.0} = 0.32$

2. **Intensity**: The 25-year intensity is taken from an intensity-duration-frequency curve for Memphis. To obtain the intensity, the time of concentration, t_c, must first be estimated. In this example, the velocity method for t_c is used to compute t_c:

Flow Path	Slope(%)	SI Unit		CU Unit	
		Length (m)	Velocity (m/s)	Length (ft)	Velocity (ft/s)
Overland (Short grass)	2	90	0.30	295	1.0
Grassed waterway	2	300	0.64	985	2.1
Grassed waterway	1	650	0.46	2,130	1.5

The time of concentration is estimated as:

Variable	Value in SI	Value in CU
$T_c = \Sigma\left(\frac{L}{V}\right)$	$= \frac{90\ m}{0.3\ m/s} + \frac{300\ m}{0.64\ m/s} + \frac{650\ m}{0.46\ m/s}$ $= 2{,}180\ s = 36\ min$	$= \frac{295\ ft}{1.0\ ft/s} + \frac{985\ ft}{2.1 ft/s} + \frac{2{,}130\ ft}{1.5\ ft/s}$ $= 2{,}180\ s = 36\ min$

The intensity is obtained from the IDF curve for the locality using a storm duration equal to the time of concentration:

$$i = 85\ mm/h\ (3.35 in/h)$$

3. **Area (A)**: Total area of drainage basin, A = 43.7 ha (108 acres)

4. **Peak Discharge (Q)**:

Variable	Value in SI	Value in CU
$Q = \frac{1}{\alpha} CiA$	$= \frac{(0.32)(85)(43.7)}{360} = 3.3\ m^3/s$	$= \frac{(0.32)(3.35)(108)}{1} = 116 ft^3/s$